

Eigenvector Methods in Quantum Systems: Adiabatic Transfer, Holographic Computation, and Interference

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Abstract

This paper presents three applications of eigenvector methods in quantum and quantum-inspired systems. First, we demonstrate adiabatic eigenvector state transfer in a three-site quantum system using counter-intuitive pulse sequences to achieve high-fidelity population transfer via a dark eigenstate. Second, we introduce a holographic prime-finding algorithm that leverages FFT-based eigenstate addition analogous to the quantum Draper adder. Third, we analyze two-qubit interference patterns arising from controlled-phase eigenvector coupling. These methods illustrate the power of eigenvector-based approaches for quantum control, computation, and interference phenomena.

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1. Introduction

Eigenvectors and eigenvalues form the mathematical foundation of quantum mechanics, where observables are represented by Hermitian operators and measurement outcomes correspond to eigenvalues. This paper explores three distinct applications of eigenvector methods: (1) adiabatic quantum state transfer, (2) holographic computation using eigenstate arithmetic, and (3) controlled-phase interference in multi-qubit systems. Each application demonstrates how eigenvector structure can be exploited for quantum control, algorithmic computation, and interference-based information processing.

2. Adiabatic Eigenvector State Transfer

2.1 Hamiltonian and Dark State

We consider a three-site quantum system with nearest-neighbor couplings:

$$H(t) = \begin{bmatrix} 0 & J_L(t) & 0 \\ J_L(t) & 0 & J_R(t) \\ 0 & J_R(t) & 0 \end{bmatrix}$$

$$|D(t)\rangle = (J_R(t)|q_0\rangle - J_L(t)|q_2\rangle) / \sqrt{J_R(t)^2 + J_L(t)^2}$$

where $|D(t)\rangle$ is the zero-energy dark eigenstate.

2.2 Counter-Intuitive Pulse Sequence

The coupling functions are Gaussian pulses with counter-intuitive timing:

$$J_L(t) = J_{\max} * \exp(-(t - T_L)^2 / (2 * \sigma^2)) \quad J_R(t) = J_{\max} * \exp(-(t - T_R)^2 / (2 * \sigma^2))$$

with $T_R < T_L$, ensuring J_R peaks before J_L .

2.3 Simulation Results

Numerical simulation of the Schrödinger equation demonstrates high-fidelity state transfer from $|q_0\rangle$ to $|q_2\rangle$ via adiabatic following of the dark eigenstate.

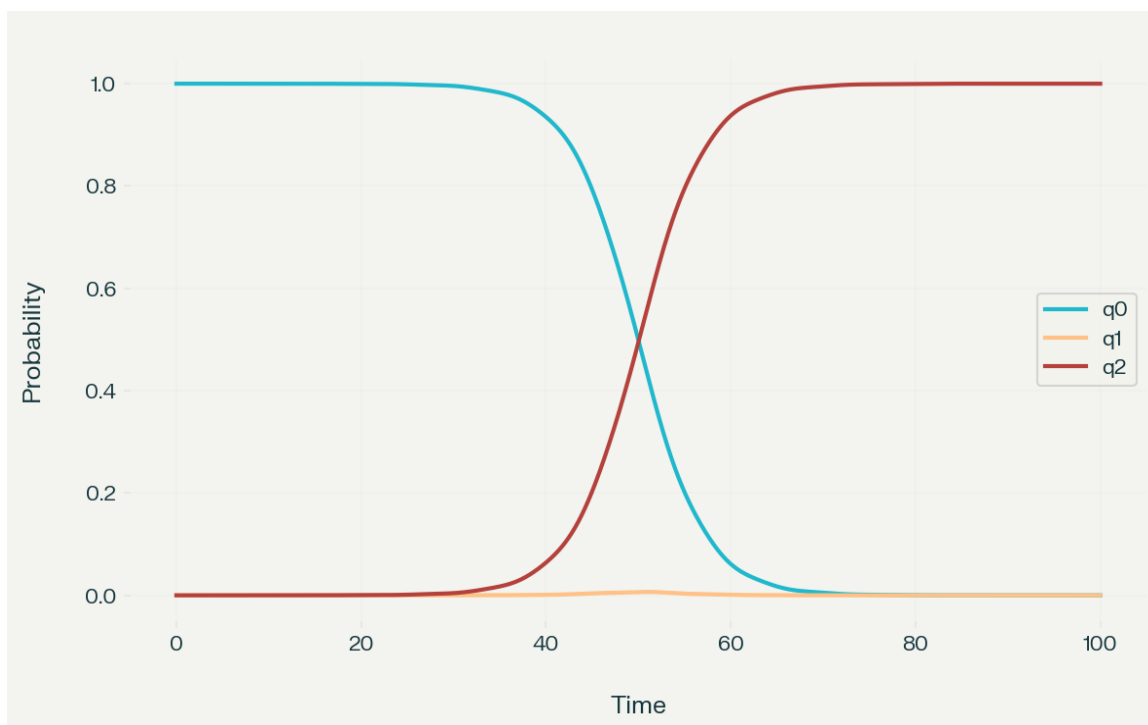


Figure 1: Adiabatic state transfer showing population dynamics.

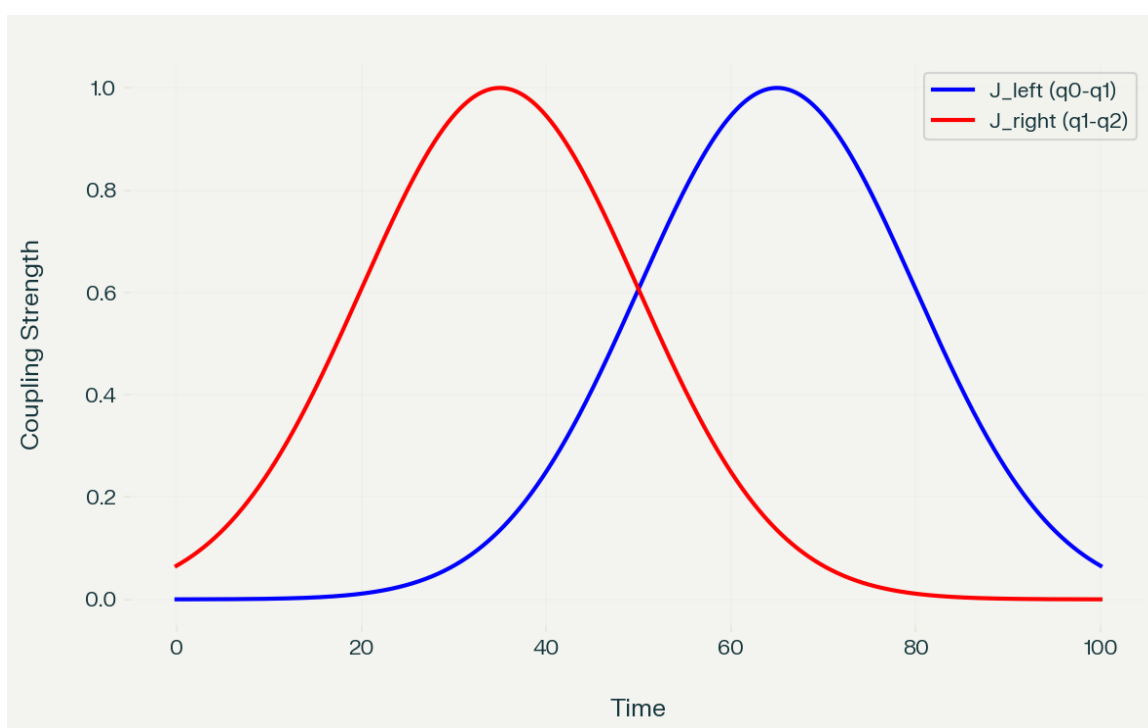


Figure 2: Counter-intuitive pulse sequence.

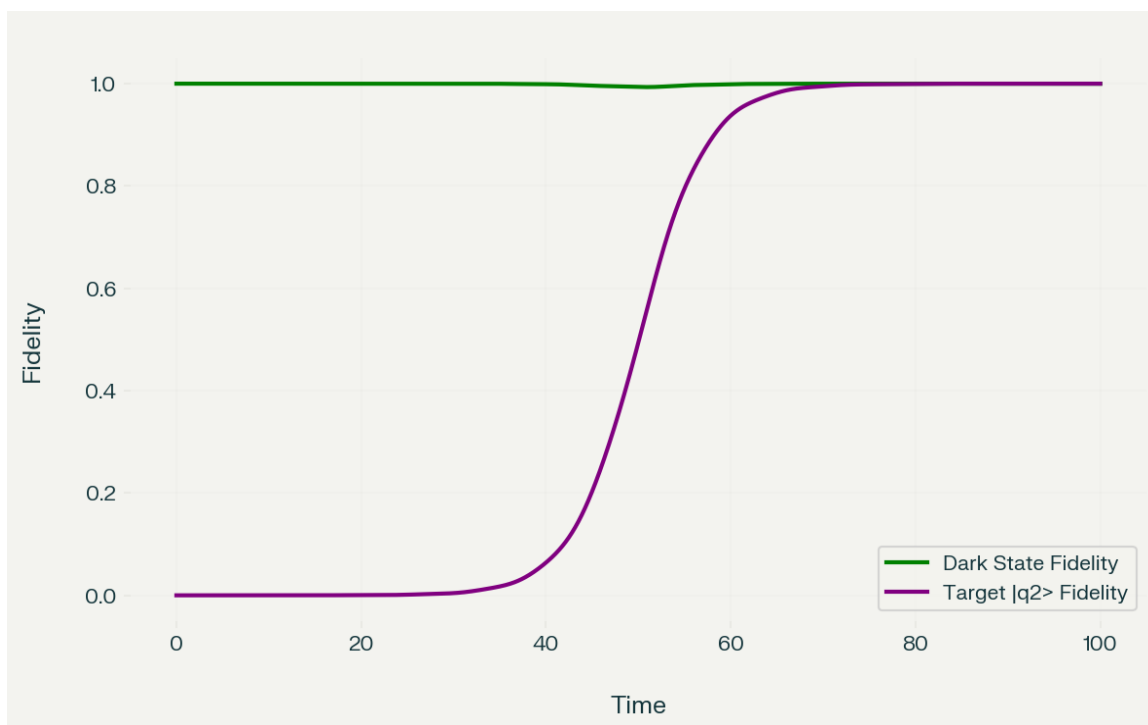


Figure 3: Fidelity with dark state and target.

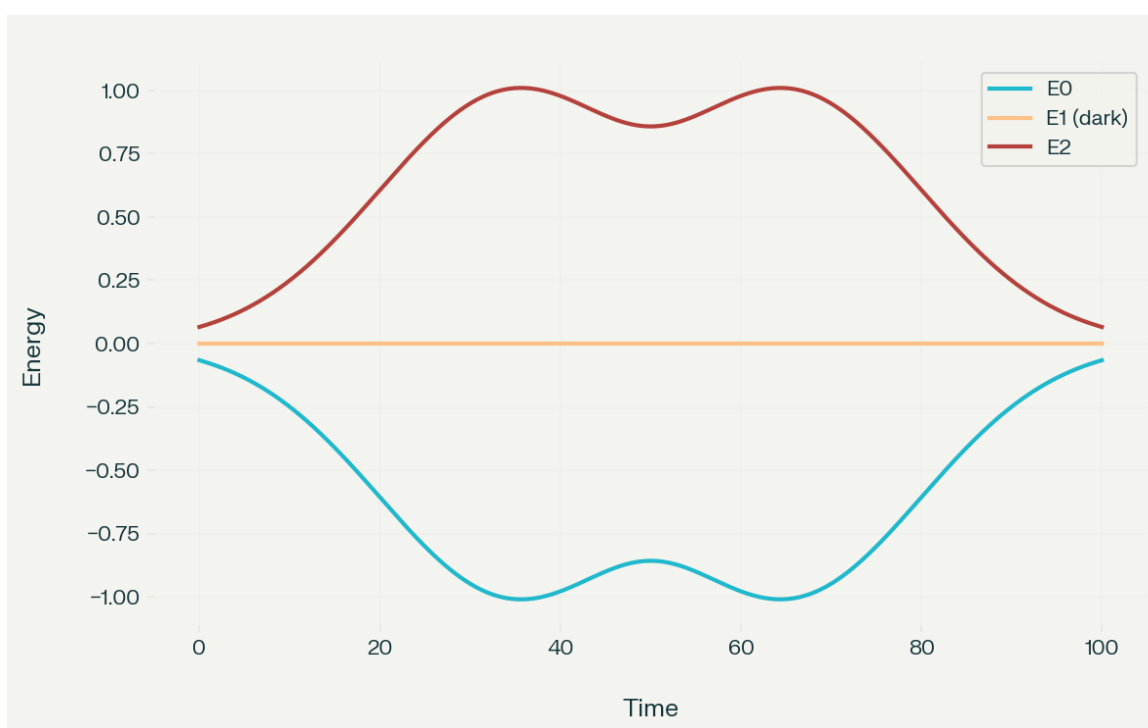


Figure 4: Instantaneous eigenvalue spectrum.

3. Holographic Prime Finding via Eigenstate Arithmetic

3.1 FFT Eigenstate Adder

The FFT eigenstate adder implements addition by moving to the eigenbasis of the cyclic shift operator:

$$v_k[x] = (1/\sqrt{n}) * \exp(2\pi i * k * x / n)$$

Addition is performed via phase shifts in the eigenbasis, analogous to the quantum Draper adder.

3.2 Prime Sieve Algorithm

The holographic prime sieve uses eigenstate-based arithmetic to test divisibility. A number n is prime if it has no divisors d where $2 \leq d \leq \sqrt{n}$.

3.3 Computational Complexity

The algorithm uses $O(\sqrt{n})$ divisibility tests, each requiring $O(\log n)$ eigenstate arithmetic operations. The overall complexity is $O(\sqrt{n} * \log n)$, comparable to classical trial division but implemented entirely through eigenstate-based operations.

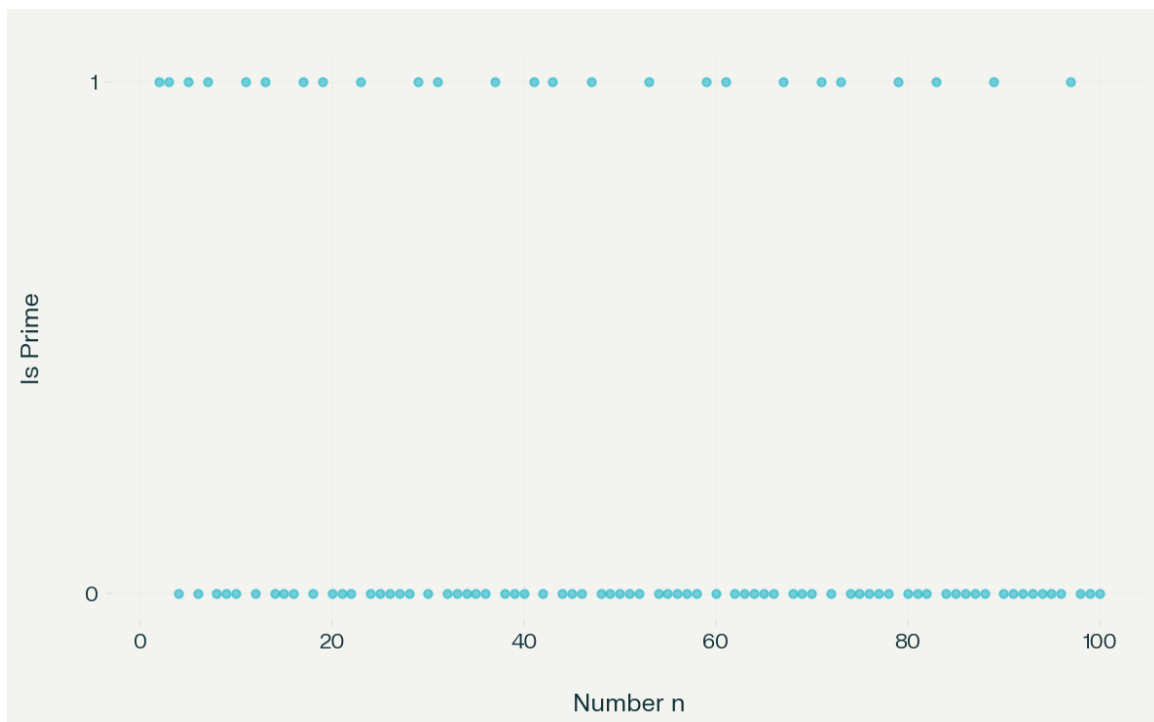


Figure 5: Distribution of primes found via eigenstate arithmetic.

4. Eigenvector-Coupled Qubit Interference

4.1 Controlled-Phase Gate

The controlled-phase gate is diagonal in the computational basis:

$$CP(\phi) = \text{diag}(1, 1, 1, \exp(i\phi))$$

The $|11\rangle$ eigenvector acquires a tunable relative phase ϕ .

4.2 Interference Pattern

A circuit of H-H-CP(ϕ)-H-H creates an interference pattern where output probabilities depend on the controlled phase ϕ .

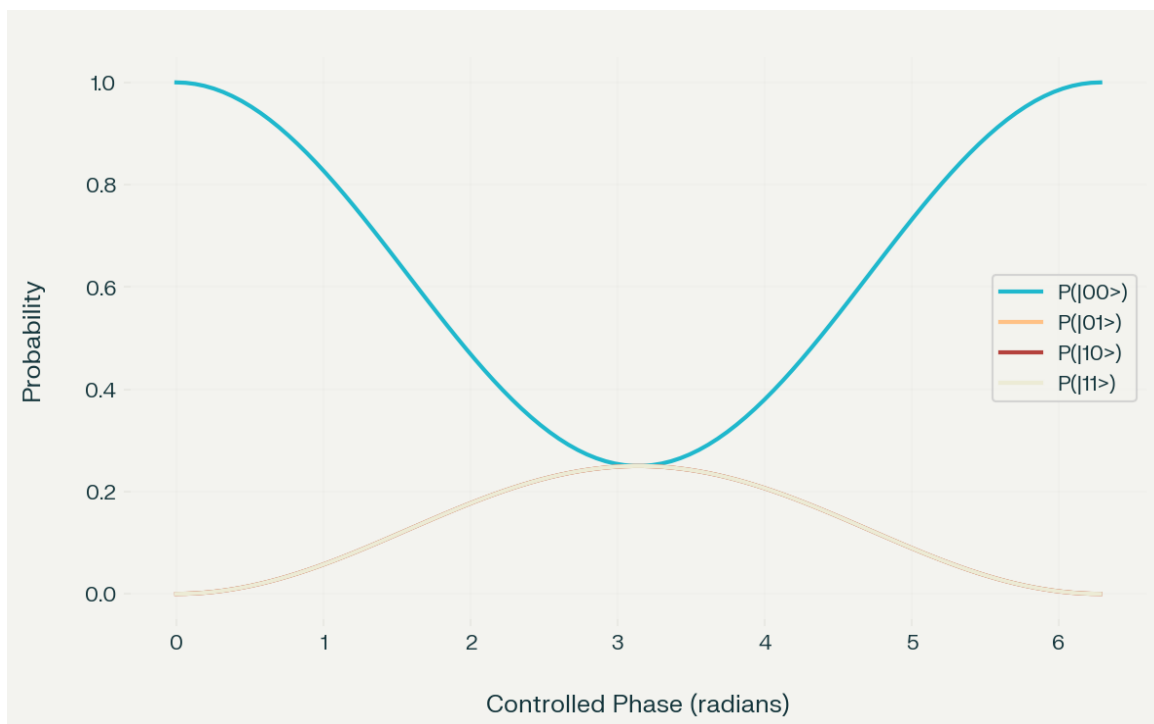


Figure 6: Two-qubit interference pattern vs controlled phase.

4.3 Entanglement and Bell States

The Bell state $|\text{Bell}\rangle = (|00\rangle + |11\rangle) / \sqrt{2}$ demonstrates entanglement where amplitudes belong to a joint two-qubit state rather than independent probability distributions.

5. Discussion and Future Directions

These three applications demonstrate the versatility of eigenvector methods across quantum control, computation, and interference. Future directions include: (1) extending adiabatic transfer to larger arrays with topological protection, (2) implementing holographic algorithms on actual quantum hardware using QFT-based adders, and (3) exploring multi-qubit interference patterns for quantum sensing and metrology applications.

6. Conclusion

Eigenvector-based methods provide powerful tools for quantum information processing. The adiabatic transfer protocol achieves high-fidelity state transfer via dark eigenstate following. The holographic prime finder demonstrates eigenstate arithmetic for algorithmic computation. The controlled-phase interference circuit illustrates how eigenvector phases create observable interference patterns. Together, these methods highlight the central role of eigenvectors in quantum science and technology.

References

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